

Geographic Data

Lecture Week 5

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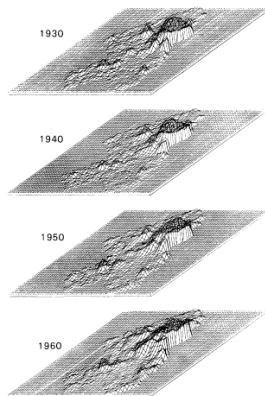
- 1 What is “distance”?
- 2 How is geography used in politics research?
- 3 Challenges of geographic data

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Tobler's Law (1970)

The First Law of Geography:

"Everything is related to everything else, but near things are more related than distant things."



Actual population growth, Detroit Region
(non-linear vertical scale).

Physical distance

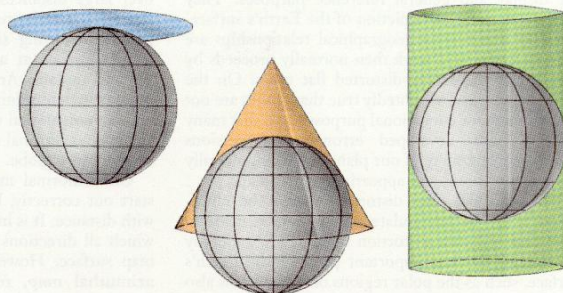
- Euclidean distance
 - Straight-line distance, based on Pythagorean Theorem
 - What does this represent? Exposure? Access?

Distance as dimension reduction



Distance as dimension reduction

FIGURE 3.2 The projection principles for plane, conic, and cylindrical map projections. To envision how the projections are accomplished, imagine a light placed at the centers of each of the globes projecting the outlines of the Earth features onto the three geometrical surfaces.



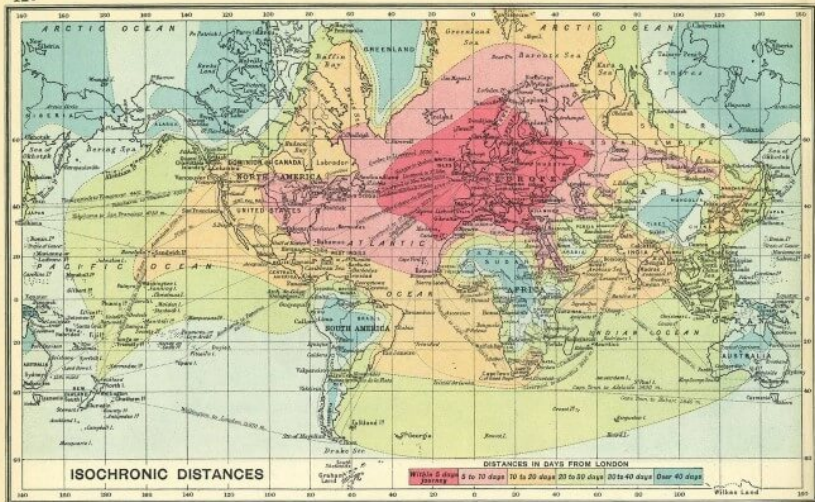
Source: <https://faculty.kutztown.edu/courtney/blackboard/Physical/05Project/project.html>

Physical distance

- Euclidean distance
 - Straight-line distance, based on Pythagorean Theorem
 - What does this represent? Exposure? Access?
- Travel distance or routing distance
 - Usually based on given mode of transit
 - **Isochrone** maps

Isochrone maps

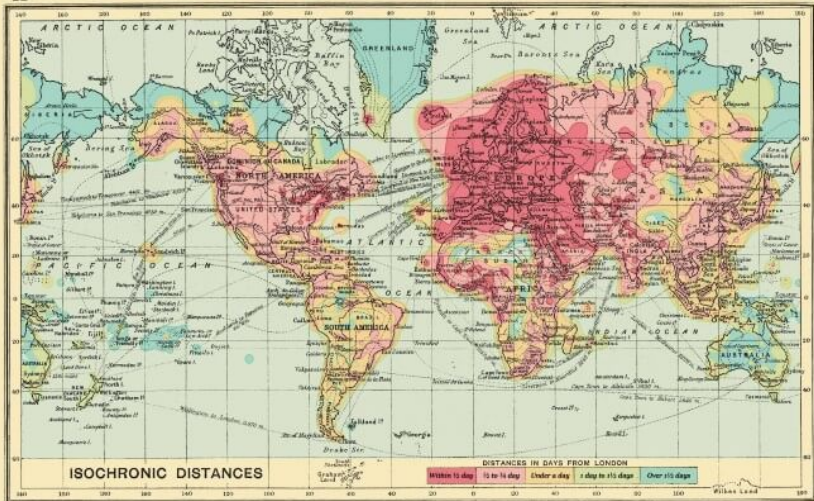
12^B



1914

Isochrone maps

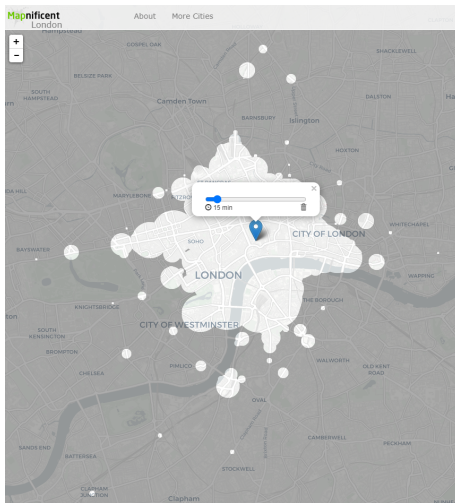
12^B



2016

HomeBio

Isochrone maps



Source: <https://www.mapnificent.net/>

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 - **Isochrone** maps → maps affect our **perceptions of space**

Maps affect perceptions



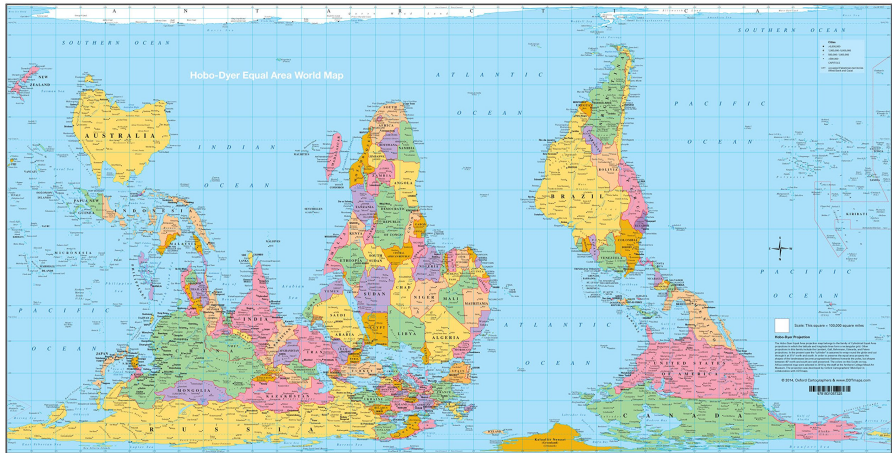
Maps affect perceptions



U.S. ARMY IN WORLD WAR II

MAP 1

Maps affect perceptions



Maps are political! (Sometimes really political)



Physical distance

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 - What does this represent? Exposure? Access?
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 - Usually based on given mode of transit
 - **Isochrone** maps → maps affect our **perceptions of space**
- Multi-dimensionality and transformations
 - Distance is a one-dimensional summary of many dimensions
 - Non-linearities in distance?

Psychological distance

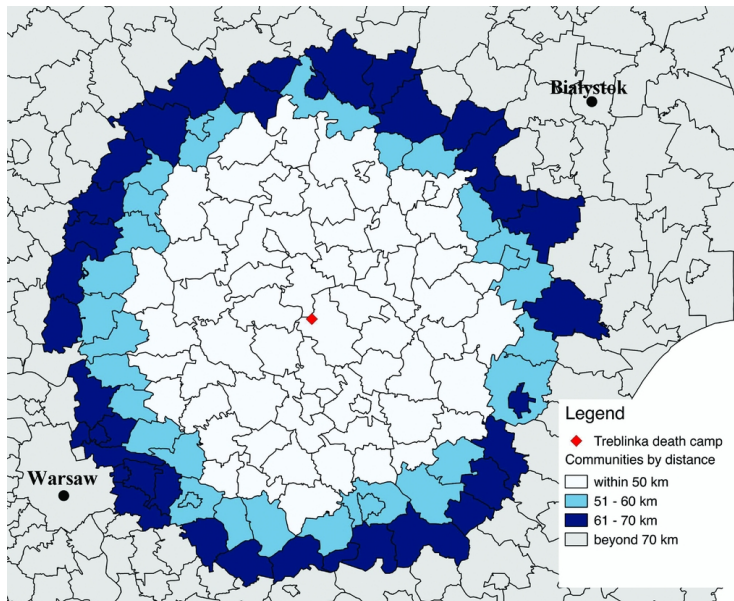
- **Psychological or cognitive distance**: “judgments we make about spatial separation of objects that cannot be seen directly” (Gatrell 1983)
 - e.g., temporal distance, spatial distance, social (or interpersonal) distance, hypothetical distance
 - Implications for language, memory, inferences we draw about other people and about ourselves, decision-making.
- **Construal Level Theory** (Trope & Liberman 2003)
 - Objects or events that are **more distant** in time or space are thought of in **more abstract** terms (higher level of construal), while those that are **closer** in time or space are thought of in **more concrete** terms (lower level of construal).
 - Goes both ways
- Political implications?
- Still relevant today?

Overview

- 1 What is “distance”?
- 2 How is geography used in politics research?
- 3 Challenges of geographic data

How is geography used in politics research?

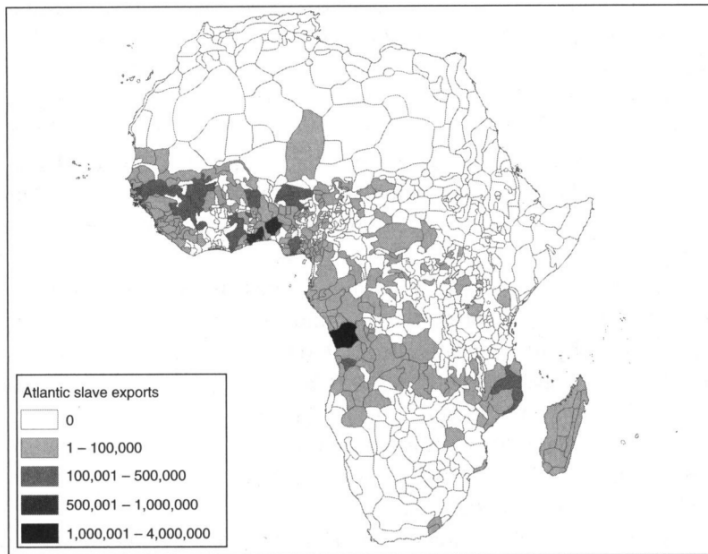
- As **explanatory** variable, e.g.,
 - Distance from Mexico-US border (Dube, Dube, & Garcia-Ponce 2013)
 - Missile range from Gaza (Getmansky & Zeitzoff 2014)
 - Proximity to Nazi death camp (Charnysh & Finkel 2017)
 - Proximity to wildfires (Hazlett & Mildemberger 2020)



How is geography used in politics research?

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- As **assignment** variable (Geo-RDD or Geo-IV), e.g.,
 - RDD: Distance to Potosi (Dell 2010)
 - RDD: Distance from colonial border (Lee & Schulz 2011)
 - IV: Distance from site of invention of written language (Laitin & Ramachandran 2016)
 - IV: Distance from coast (Nunn & Wantchekon 2011)
 - IV: Distance from railway network (Rozenas, Schutte, & Zhukov 2017)

Panel A. Transatlantic slave trade



How is geography used in politics research?

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- As mediating variable, e.g.,
 - Distance to polling station (Brady & McNulty 2011)
- As control variable

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Challenges of geographic data

1 Operationalisation and measurement

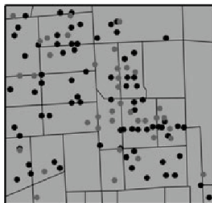
- Distance? Exposure? Geographic unit?

2 Modifiable Areal Unit Problem (MAUP)

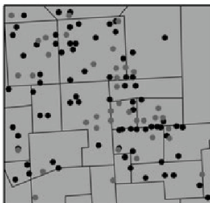
- Summary values are influenced by both the **shape** and **scale** of the aggregation unit

Modifiable Areal Unit Problem

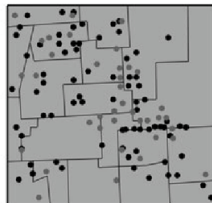
Census tracts



Voting districts

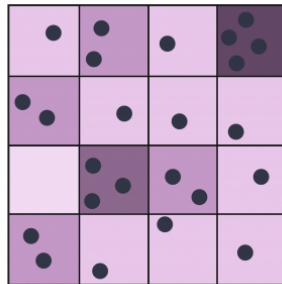
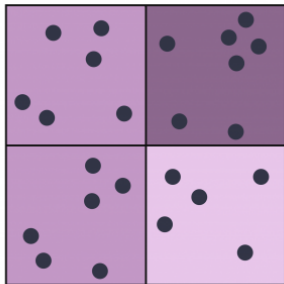


Police precincts



● Armed robbery

● Assaults



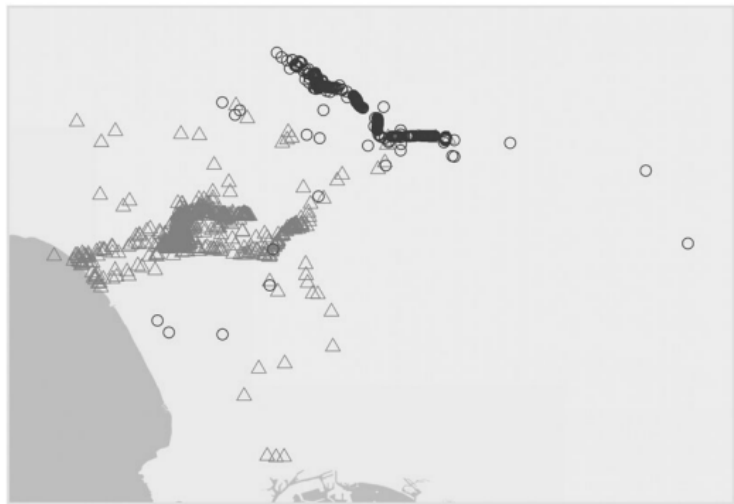


Figure 1. The paths of two individuals in Los Angeles County. Person 1 (Δ) is concentrated in West Los Angeles and Person 2 (○) in La Crescenta-Montrose.

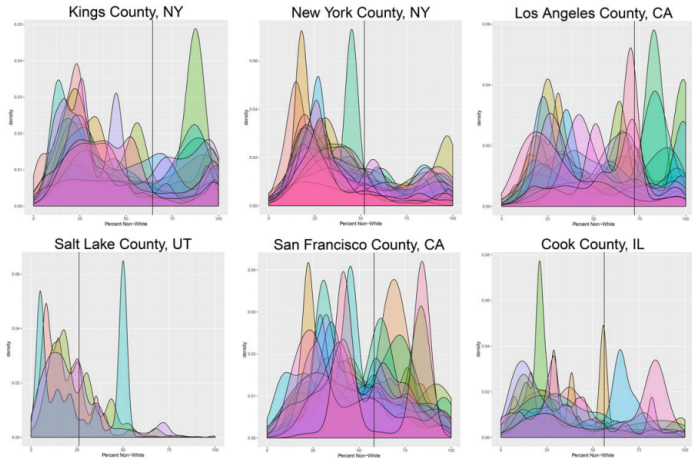
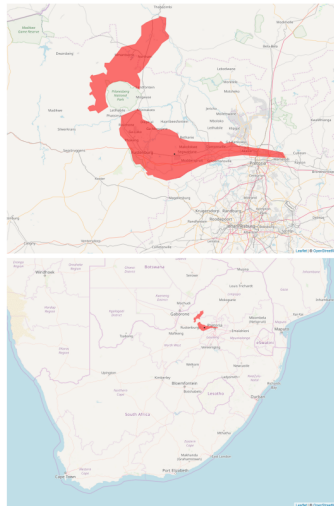


Figure 5. Examples of racial experiences in select metropolitan counties. Individuals living in the same county have disparate racial experiences. Panels show the distribution of the racial and ethnic composition of the census blocks that individuals travel through, grouped by modal county. The x-axis is the percent non-white; densities farther to the right indicate more diverse (less white) racial and ethnic contexts. Vertical lines represent the percent of the county population that is non-white.

Challenges of geographic data

- ❶ Operationalisation and measurement
 - Distance? Exposure? Geographic unit?
- ❷ Modifiable Areal Unit Problem (MAUP)
 - Summary values are influenced by both the shape and scale of the aggregation unit
- ❸ Measurement and data problems
 - non-matching CRS and projections
 - polygons vs. points; centroids
 - unit boundaries don't match, or change over time
- ❹ Distance/geography for causal inference
 - Geo-IVs (plausibility? Tobler...)
 - Geo-RDD (see Keele & Titiunik, 2015)
 - Spatial diff-in-diff

FIGURE 1. The Western Limb and the Kopple, the Site of the Marikana Massacre



Note: The upper panel shows the site of the Marikana Massacre (the black dot) and the extent of the Western Limb communities (red polygons) in the local geographic context. The lower panel shows the same but in a broader context. The Western Limb polygons were hand drawn by the authors in GIS, following Godfrey (2018) and Kinnaird (2008). The massacre occurred at two different scenes; the black dot shows the Kopple from which the strike was organized, which lies equidistant between them, 250m from either scene.

